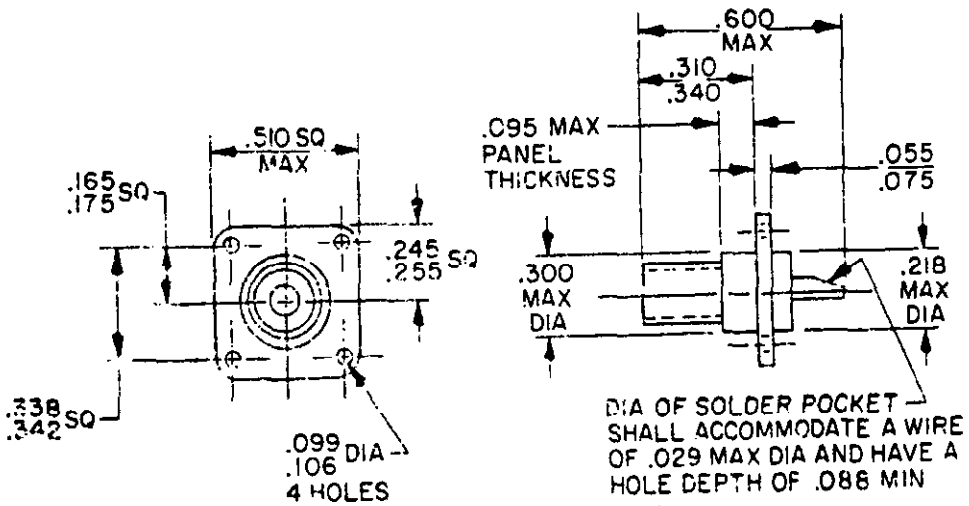
MILITARY SPECIFICATION SHEET

CONNECTORS, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA (UNCABLED, SOCKET CONTACT, FLANGE MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-39012.



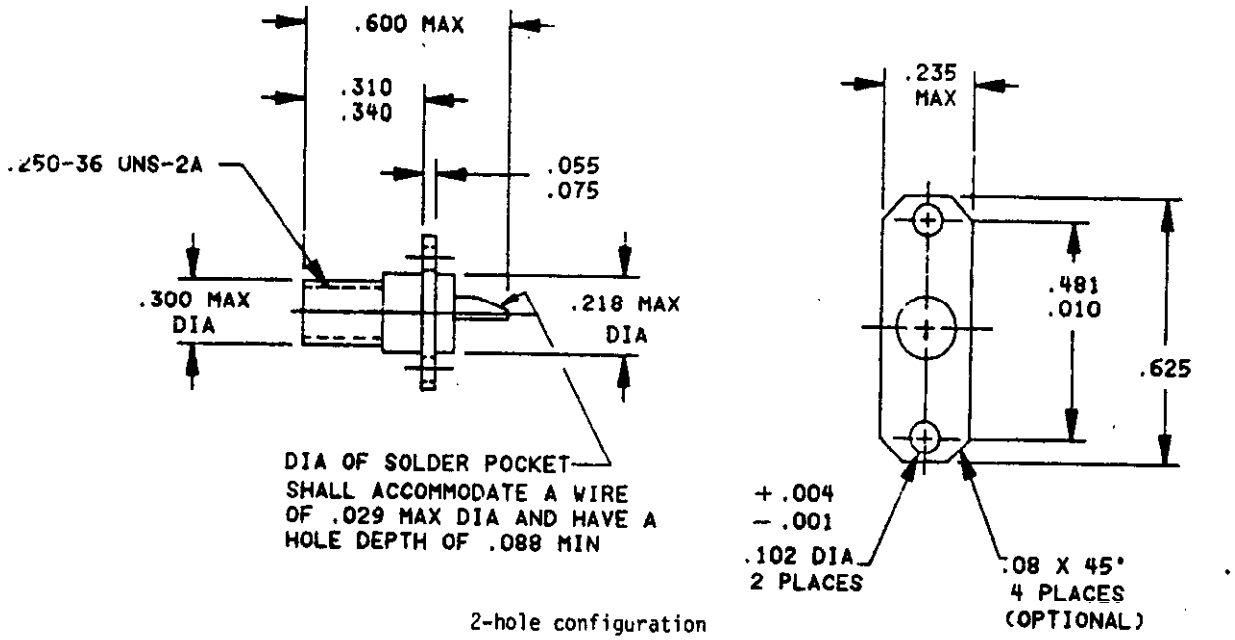
INCHES	MM	INCHES	MM
.029	0.74	.245	6.22
.055	1.40	.255	6.48
.075	1.91	.300	7.62
.086	2.24	.310	7.87
.095	2.41	.338	8.49
.099	2.51	.340	8.64
.106	2.69	.342	8.69
.165	4.19	.510	12.95
.175	4.45	.600	15.24
.218	5.54		

4-hole configuration
MS39012/60-3001
MS39012/60-4001

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Receptacle recommended for component part use and for use on panels with a .080 (2.03 mm) maximum thickness.
4. All undimensioned pictorial configurations are for reference purposes only.
5. There shall be a solid barrier in the socket between the pin entry and the solder pocket to prevent solder wicking.
6. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on three place decimals, and ±.010 (0.25 mm) on two place decimals, and ±2° on angles.

FIGURE 1. General configuration.

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2-hole configuration

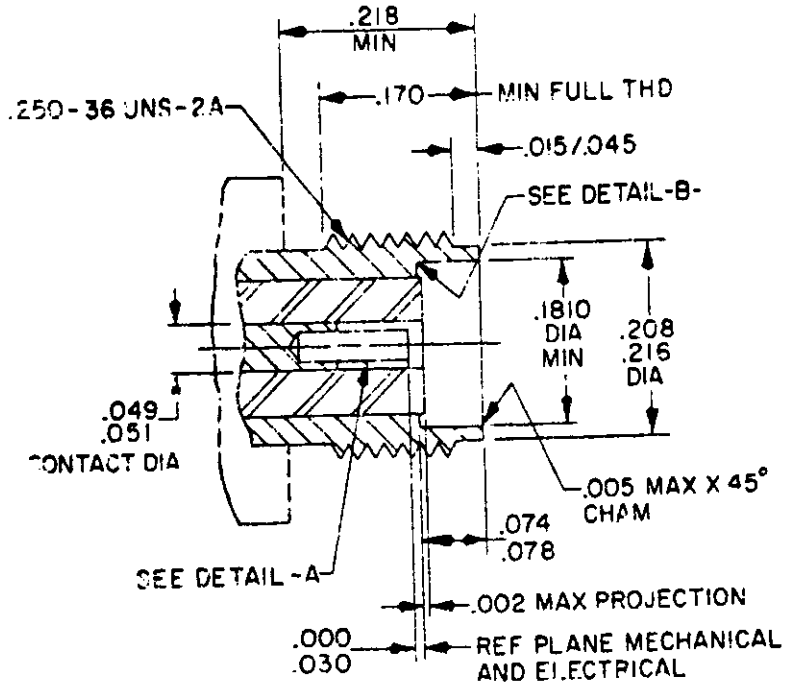
MS39012/60-3002
MS39012/60-4002

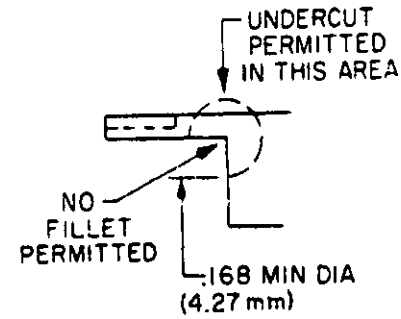
INCHES	MM	INCHES	MM
.029	0.74	.218	5.54
.055	1.40	.300	7.62
.075	1.91	.310	7.87
.088	2.24	.340	8.64
.095	2.41	.600	15.24

INCHES	MM	INCHES	MM
.001	0.02	.102	2.59
.004	0.10	.233	5.92
.010	0.25	.481	12.22
.080	2.03	.625	15.88

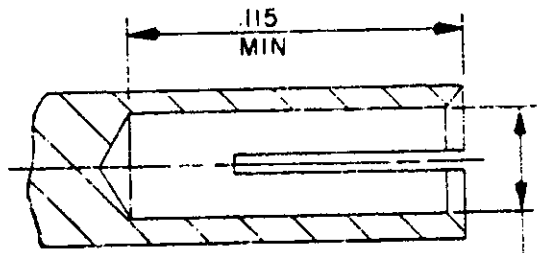
FIGURE 1. General configuration - Continued.

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DETAIL-B-



DETAIL-A-

INCHES	MM
.002	0.05
.003	0.08
.005	0.13
.015	0.38
.030	0.76
.0355	0.902
.0370	0.940
.045	1.14
.049	1.24
.051	1.30
.074	1.88
.078	1.98
.115	2.92
.170	4.32
.1810	4.597
.208	5.28
.216	5.49
.218	5.54

NOTES:
1. Dimensions are in inches.
2. Slitting of inner contact optional.
3. Metric equivalents are given for general information only.

FIGURE 2. Mating dimensions for female termination.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: (Not rated.)

Voltage rating: 335 V rms maximum at sea level. 85 V rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force - Not applicable.
Torque - 2 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics (see figures 2 and 3 for dimensions):

Contacts with spring members:

Center contact (female):

Oversize test pin - .0375 + .0001.

Test pin finish - 16 MICROINCHES.

Insertion depth - .030/.045.

Number of insertions - 3.

Insertion force test: Steel test pin diameter .0370 + .0001.

Insertion depth - .050/.075.

Test pin finish - 16 MICROINCHES.

Insertion force - 3 lbs maximum.

Withdrawal force test: Steel test pin diameter .0355 - .0001.

Insertion depth - .050/.075.

Withdrawal force - 1 ounce minimum.

Test pin finish - 16 MICROINCHES

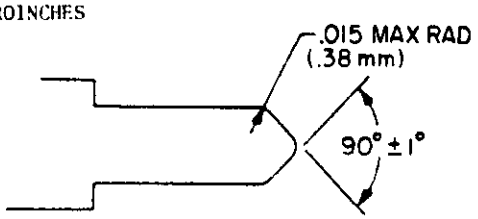


FIGURE 3. Test pin data.

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Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 3003, test condition B, MIL-STD-1344. 5,000 megohms minimum.

Center contact retention: 6 lbs minimum axial force. 4 inch-ounces minimum radial torque.

Solderability: Method 208 of MIL-STD-202. 1/

Resistance to test prod damage: Not applicable.

Corrosion (salt spray): Method 1001, test condition B, MIL-STD-1344.

Voltage standing wave ratio (VSWR): Not applicable.

Swept frequency VSWR test setup: Not applicable.

Connector durability: Method 2016, MIL-STD-1344. 500 cycles minimum at 12 cycles/min. maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact:	3.0	4.0
Outer contact:	2.0	Not applicable

Dielectric withstanding voltage at sea level: Method 3001, MIL-STD-1344. 1,000 V rms.

Vibration high frequency: Method 2005, test condition IV, MIL-STD-1344.

Shock: Method 2004, test condition G, MIL-STD-1344.

Thermal shock: Method 1003, test condition A, MIL-STD-1344, except test high temperature to be +200°C.

Humidity: Method 1002 of MIL-STD-1344. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 250 volts minimum.
Altitude - 70,000 feet.

RF high potential withstanding voltage:

Frequency - 670 V rms at 5 MHz.
Leakage current - Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: Not applicable.

RF insertion loss: Not applicable.

1/ For quality conformance inspection, the test shall be performed in group B following VSWR; inspection level shall be S-1 of MIL-STD-105 and AQL 10 percent defective.

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TABLE I. Group qualification and retention testing.

Group	Submission and qualification of any one of the following connectors 1/ 2/	Qualifies the following connectors
I	M39012/61-3001 M39012/61-4001 M39012/61-3002 M39012/61-4002 M39012/60-3001 M39012/60-4001 M39012/60-3002 M39012/60-4002	M39012/61-3001 M39012/61-4001 M39012/61-3002 M39012/61-4002 M39012/60-3001 M39012/60-4001 M39012/60-3002 M39012/60-4002

1/ Qualification of connectors qualifies connectors of the same material only.

2/ For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column. The part does not necessarily have to be the part initially qualified.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AM, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3428)

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